Monte Carlo Pricing of a European Call: Control Variates

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Abstract

We apply the control variates technique of the note on variance-reduction foundations to the Monte Carlo pricer of the European call under the exact GBM sampler. Two controls are introduced: the discounted underlying $X_1 = e^{-rT} S_T$ and the discounted asset-or-nothing payoff $X_2 = e^{-rT} S_T \mathbf{1}_{\{S_T > K\}}$. Both have closed-form expectations under GBM. *A naive shape-matching argument suggests X_2 should outperform X_1 because it matches the call's payoff structurally on both the ITM and OTM regions; this prediction is empirically wrong.* The correct analysis, carried out in detail below, reveals that X_2 carries excess variance unrelated to the target and ends up with lower Pearson correlation than X_1 (typical values: $\rho_1 \approx 0.92$, $\rho_2 \approx 0.77$). The underlying control delivers VRF ≈ 6.9 at the ATM point; the AON control delivers only VRF ≈ 2.5. The lesson generalises beyond this example: structural similarity does not imply linear correlation.

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1 Introduction

The note on variance-reduction foundations derives the control-variate framework: given a target Y with unknown $\mathbb{E}[Y]$ and a control X with known $\mathbb{E}[X]$ correlated with Y, the adjusted estimator

$$\tilde{Y}(c) \;=\; Y - c\,(X - \mathbb{E}[X]) \tag{1}$$

is unbiased for $\mathbb{E}[Y]$ for any choice of c, with variance

$$\mathrm{Var}(\tilde{Y}(c^*)) \;=\; \mathrm{Var}(Y)(1-\rho^2), \quad c^* \;=\; \frac{\mathrm{Cov}(Y,X)}{\mathrm{Var}(X)}, \quad \rho \;=\; \mathrm{Corr}(Y,X). \tag{2}$$

The variance reduction factor is $\mathrm{VRF} = 1/(1-\rho^2)$, which can be very large when $|\rho|$ is close to 1.

The challenge in applying CV is finding a control. It must satisfy two conditions: (a) its expectation must be known in closed form, (b) it must be highly correlated with the target. The first is restrictive; the second is what distinguishes a good control from a mediocre one.

For the European call under GBM, two controls present themselves naturally: the discounted underlying $X_1 = e^{-rT}S_T$ and the discounted asset-or-nothing payoff $X_2 = e^{-rT}S_T\mathbf{1}_{\{S_T>K\}}$. Both are tractable in closed form. The expectations of these two controls present a clean opportunity to illustrate a subtle but important pedagogical point about how correlation works, which is the main content of this writeup.

2 Control 1: the discounted underlying

2.1 Definition and tractability

The discounted underlying at maturity is

$$X_1 \;\stackrel{\text{def}}{=}\; e^{-rT}S_T. \tag{3}$$

Under the risk-neutral measure $\mathbb{Q}$, $\{e^{-rt}S_t\}_{t\geq 0}$ is a martingale, so

$$\mathbb{E}^{\mathbb{Q}}[X_1] \;=\; e^{-rT}\mathbb{E}^{\mathbb{Q}}[S_T] \;=\; e^{-rT}\cdot S_0 e^{rT} \;=\; S_0. \tag{4}$$

The expectation is therefore known exactly, with no computation needed. This is the simplest possible control variate for any GBM-based pricer.

2.2 Naive correlation prediction

The relationship between $Y = e^{-rT}(S_T - K)^+$ and $X_1 = e^{-rT}S_T$ is:

- On $\{S_T > K\}$ (ITM): $Y = X_1 - e^{-rT}K$, linear in X_1 with slope 1.
- On $\{S_T \leq K\}$ (OTM): $Y = 0$, while $X_1 > 0$. Here Y is constant and uncorrelated with X_1 locally.

A naive prediction based on this dichotomy is that ρ_1 should be moderate, perhaps 0.5–0.7, since one of the two regimes (OTM) "wastes" the linear relationship. We will see below that this prediction *underestimates* ρ_1 substantially: the actual value at the ATM point is $\rho_1 \approx 0.92$.

3 Control 2: the asset-or-nothing payoff

3.1 Definition and tractability

The discounted asset-or-nothing call payoff is

$$X_2 \;\stackrel{\text{def}}{=}\; e^{-rT}\, S_T \mathbf{1}_{\{S_T > K\}}. \tag{5}$$

Its expectation has a closed Black-Scholes form. From the standard derivation of the BS call price decomposition,

$$\mathbb{E}^{\mathbb{Q}}[X_2] \;=\; e^{-rT}\mathbb{E}^{\mathbb{Q}}\big[S_T \mathbf{1}_{\{S_T > K\}}\big] \;=\; S_0\, \Phi(d_1), \tag{6}$$

where

$$d_1 \;=\; \frac{\log(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \tag{7}$$

is the standard d_1 from Black-Scholes. This identity is the asset-or-nothing building block of the BS formula

$$C^{\text{BS}} \;=\; S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2), \quad d_2 = d_1 - \sigma\sqrt{T}, \tag{8}$$

so $\mathbb{E}[X_2]$ is available for free as a byproduct of the BS pricer code in `quantlib.black_scholes`.

3.2 The naive intuition: "shape matching"

The relationship between Y and X_2 is:

- On $\{S_T > K\}$: $Y = X_2 - e^{-rT}K$, linear in X_2 with slope 1.
- On $\{S_T \le K\}$: $Y = 0$ *and* $X_2 = 0$ simultaneously, both at the origin.

The naive intuition is that X_2 should outperform X_1 because it matches the target on *both* regions: zero where the target is zero, and linearly related where the target is linear. The OTM "wasted region" that hurt X_1 does not exist for X_2; instead both target and control vanish together. One might therefore predict ρ_2 near 1, perhaps above 0.95.

This prediction is wrong, and the rest of this writeup is devoted to understanding why. Structural similarity — vanishing together, linear together — is not the same as global linear correlation, and the difference is exactly what Pearson's ρ measures.

4 The empirical surprise and the correct analysis

We computed ρ_1 and ρ_2 analytically (and confirmed empirically with 10^6 paths) for the canonical ATM parameters ($S_0 = K = 100, r = 0.05, \sigma = 0.20, T = 1$). The results:

$$\rho_1 \;=\; 0.9245, \qquad \rho_2 \;=\; 0.7714, \tag{9}$$

$$\text{VRF}_1 \;=\; \frac{1}{1-\rho_1^2} \;=\; 6.88, \qquad \text{VRF}_2 \;=\; \frac{1}{1-\rho_2^2} \;=\; 2.47. \tag{10}$$

The underlying control beats the asset-or-nothing control by almost a factor of 3 in variance reduction. The naive prediction was inverted: X_1 is the better control, not X_2.

4.1 The closed-form analysis

The key algebraic observation is that Y and X_2 differ by a deterministic function of the path:

$$Y \;=\; X_2 - e^{-rT}K \cdot A, \qquad A \;\stackrel{\text{def}}{=}\; \mathbf{1}_{\{S_T > K\}}. \tag{11}$$

On $\{S_T > K\}$ both sides equal $e^{-rT}(S_T - K)$; on $\{S_T \le K\}$ both sides equal zero. So Y is exactly X_2 shifted down by a Bernoulli random variable scaled by the discounted strike.

From this identity:

$$\mathrm{Var}(Y) \;=\; \mathrm{Var}(X_2) - 2e^{-rT}K\ \mathrm{Cov}(X_2, A) + (e^{-rT}K)^2\,\mathrm{Var}(A), \tag{12}$$

$$\mathrm{Cov}(Y, X_2) \;=\; \mathrm{Var}(X_2) - e^{-rT}K\ \mathrm{Cov}(X_2, A). \tag{13}$$

The closed forms for the second moments under GBM (lognormal MGF applied to the truncated regions) are:

$$\mathbb{E}[X_1] = S_0, \qquad \mathrm{Var}(X_1) = S_0^2\big(e^{\sigma^2 T} - 1\big), \tag{14}$$

$$\mathbb{E}[X_2] = S_0\Phi(d_1), \qquad \mathbb{E}[X_2^2] = S_0^2\, e^{\sigma^2 T}\,\Phi(d_1 + \sigma\sqrt{T}), \tag{15}$$

and the cross terms $\mathrm{Cov}(X_1, A)$, $\mathrm{Cov}(X_2, A)$, $\mathrm{Cov}(X_1, X_2)$ likewise have closed forms in terms of $\Phi(d_1)$, $\Phi(d_2)$, and $\Phi(d_1 + \sigma\sqrt{T})$. We omit the routine algebra and quote the numerical values for our parameters:

$$\mathrm{Var}(Y) = 216.66, \quad \mathrm{Var}(X_1) = 408.11, \quad \mathrm{Var}(X_2) = 3322.15. \tag{16}$$

The most informative comparison is between $\mathrm{Var}(X_1)$ and $\mathrm{Var}(X_2)$. The variance of the asset-or-nothing control is *eight times* that of the underlying control, and *fifteen times* that of the target itself.

4.2 Why excess variance hurts correlation

Pearson's correlation is

$$\rho \;=\; \frac{\mathrm{Cov}(Y, X)}{\sqrt{\mathrm{Var}(Y)\cdot\mathrm{Var}(X)}}. \tag{17}$$

The covariance and the target variance are fixed by the problem; the control's variance enters in the denominator. *A control with excess variance — variance not explained by linear association with Y — gets penalised in the correlation coefficient.*

In our case the excess variance of X_2 does not come from the range of S_T on the ITM region: conditional on $\{S_T > K\}$, the variables Y, X_1 and X_2 differ only by constants and so have the same variance, and on OTM both Y and X_2 vanish. It comes from the jump of height $e^{-rT}K$ that X_2 makes at $S_T = K$, where Y is continuous. Writing (11) as $X_2 = Y + e^{-rT}K\,A$,

$$\mathrm{Var}(X_2) - \mathrm{Var}(Y) \;=\; (e^{-rT}K)^2\,\mathrm{Var}(A) + 2e^{-rT}K\ \mathrm{Cov}(Y, A) \;=\; 2229.9 + 875.6 \;=\; 3105.5, \tag{18}$$

with $\mathrm{Var}(A) = \Phi(d_2)\big(1 - \Phi(d_2)\big)$ and $\mathrm{Cov}(Y, A) = C^{\mathrm{BS}}\big(1 - \Phi(d_2)\big)$; this recovers $3322.15 - 216.66$. The denominator of ρ_2 therefore picks up the full $\mathrm{Var}(X_2)$ while the numerator $\mathrm{Cov}(Y, X_2)$ picks up only the portion correlated with Y.

For X_1, the analogous algebra plays out differently. X_1 has *positive but limited* variance on both ITM and OTM regions: the variance on OTM is constrained because $S_T \in (0, K]$ there. This bounded behaviour makes $\mathrm{Var}(X_1)$ small, only $1.9 \times \mathrm{Var}(Y)$, so the denominator does not penalise the correlation heavily.

4.3 The optimal coefficient confirms the picture

Another way to see the asymmetry is via the optimal coefficient c^*:

$$c_1^* = \frac{\mathrm{Cov}(Y, X_1)}{\mathrm{Var}(X_1)} = \frac{274.91}{408.11} = 0.674, \tag{19}$$

$$c_2^* = \frac{\mathrm{Cov}(Y, X_2)}{\mathrm{Var}(X_2)} = \frac{654.44}{3322.15} = 0.197. \tag{20}$$

For the underlying, $c_1^* \approx 0.67$: subtracting $0.67 \times (X_1 - S_0)$ from Y removes a fraction $\rho_1^2 \approx 0.855$ of $\mathrm{Var}(Y)$. For the AON, $c_2^* \approx 0.20$, much smaller: the algorithm "does not trust" the AON nearly as much as it does the underlying, because the AON has so much variance unrelated to Y.

5 The general lesson

The mistake corrected in Section 4 is a subtle one. The naive intuition — that X_2 matches Y on both regions — is correct: Y and X_2 are zero together and linear together. But this kind of *piecewise* matching is not what Pearson's correlation measures; Pearson's correlation measures *global linear fit*. Two variables can match each other piecewise and still have moderate correlation if the piecewise pieces have very different variances.

The ratio $\mathrm{Var}(Y)/\mathrm{Var}(X)$ is not a usable diagnostic for this: replacing the control X by aX with $a > 0$ leaves ρ and the CV estimator unchanged (c^* becomes c^*/a) but divides the ratio by a^2. What decides the quality of a control is the correlation itself, which here is available in closed form from the moments of Section 4. The naive shape-matching argument does not predict it.

6 Implementation: empirical $\hat{c}$

We use the empirical estimator of c^* from the same paths that produce the estimator. Given samples $\{(Y_i, X_i)\}_{i=1}^M$:

$$\hat{c} = \frac{\sum_{i=1}^M (Y_i - \bar{Y})(X_i - \bar{X})}{\sum_{i=1}^M (X_i - \bar{X})^2} \quad \text{(OLS slope of } Y \text{ on } X\text{)}, \tag{21}$$

$$\hat{C}_M^{\mathrm{CV}} = \frac{1}{M} \sum_{i=1}^M \big(Y_i - \hat{c}\,(X_i - \mathbb{E}[X])\big). \tag{22}$$

The bias of $\hat{C}_M^{\mathrm{CV}}$ is $O(1/M)$, while the standard error is $O(1/\sqrt{M})$, so the bias is asymptotically negligible for any M at which the standard error is acceptable. See the note on variance-reduction foundations, Proposition 4.4.

6.1 Connection with OLS regression

The slope estimator $\hat{c}$ is exactly the slope of the OLS regression of Y on X with an intercept. The CV-adjusted samples $\tilde{Y}_i = Y_i - \hat{c}(X_i - \mathbb{E}[X])$ are the regression residuals shifted by $\hat{\alpha} + \hat{c}\mathbb{E}[X]$ to recover the original mean. The variance of $\tilde{Y}_i$ is the variance of the regression residuals, which is $\mathrm{Var}(Y)(1 - R^2)$ with $R^2 = \rho^2$ for a single regressor. The CV variance reduction is therefore exactly the OLS coefficient of determination.

7 Algorithmic specification

Algorithm 1 Control-variate Monte Carlo for the European call (exact GBM)

Require: Spot S_0, strike K, maturity T, rate r, volatility σ, sample size M, control identifier $c \in \{1, 2\}$, confidence level $1 - \beta$, random generator `rng`.
Ensure: Estimate $\hat{C}^{\mathrm{CV}}$, half-width h_M^{CV}, sample variance $S_M^{2,\mathrm{CV}}$.

$D \leftarrow e^{-rT}$
for $i = 1, \ldots, M$ **do**
  Draw $Z_i \sim \mathcal{N}(0, 1)$ using `rng`
  $S_T^{(i)} \leftarrow S_0 \exp((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z_i)$
  $Y_i \leftarrow D \cdot \max(S_T^{(i)} - K, 0)$
  if $c = 1$ ▷ control: discounted underlying **then**
    $X_i \leftarrow D \cdot S_T^{(i)}, \quad \mathbb{E}X \leftarrow S_0$
  else if $c = 2$ ▷ control: asset-or-nothing **then**
    $X_i \leftarrow D \cdot S_T^{(i)} \cdot \mathbf{1}_{\{S_T^{(i)} > K\}}$
    $\mathbb{E}X \leftarrow S_0\, \Phi(d_1)$ ▷ closed BS form
  end if
end for
$\bar{Y} \leftarrow M^{-1} \sum_i Y_i, \quad \bar{X} \leftarrow M^{-1} \sum_i X_i$
$\hat{c} \leftarrow \frac{\sum_i (Y_i - \bar{Y})(X_i - \bar{X})}{\sum_i (X_i - \bar{X})^2}$ ▷ OLS slope
for $i = 1, \ldots, M$ **do**
  $\tilde{Y}_i \leftarrow Y_i - \hat{c}\,(X_i - \mathbb{E}X)$
end for
Reduce: $\hat{C}^{\mathrm{CV}} \leftarrow M^{-1} \sum_i \tilde{Y}_i, \quad S_M^{2,\mathrm{CV}} \leftarrow (M-1)^{-1} \sum_i (\tilde{Y}_i - \hat{C}^{\mathrm{CV}})^2$
$h_M^{\mathrm{CV}} \leftarrow z_{\beta/2} \sqrt{S_M^{2,\mathrm{CV}}/M}$
return $(\hat{C}^{\mathrm{CV}}, h_M^{\mathrm{CV}}, S_M^{2,\mathrm{CV}})$

8 Empirical validation strategy

The validation script for the CV pricers comprises four tests, run separately for each control.

Test 1: empirical correlation and VRF. For each control, generate 10^6 paths and measure $\hat{\rho} = \widehat{\mathrm{Corr}}(Y, X)$ and $\widehat{\mathrm{VRF}} = 1/(1 - \hat{\rho}^2)$. The *corrected* predictions, based on the analysis of Section 4:

- Control 1 (underlying): $\hat{\rho}_1 \approx 0.92$, $\mathrm{VRF}_1 \approx 6.9$.
- Control 2 (AON): $\hat{\rho}_2 \approx 0.77$, $\mathrm{VRF}_2 \approx 2.5$.

The control 1 (underlying) is the better choice for European call pricing under GBM, contrary to the naive shape-matching prediction.

Test 2: BS coherence. Each pricer at $M = 50\,000$ paths should agree with C^{BS} within a few half-widths.

Test 3: half-width comparison vs IID at fixed budget. For each control, run the CV pricer with $M = 100\,000$ paths and the IID pricer with the same $M = 100\,000$ paths from the same seed. The squared half-width ratios IID/CV should match the VRFs from Test 1.

Test 4: input validation. Mechanical: both pricers reject bad inputs and accept negative r.

9 Bibliographic notes

The textbook reference is Glasserman [2, Section 4.1]. The asset-or-nothing decomposition of the BS formula is in Hull [3, Chapter 26]. The OLS interpretation of the optimal control coefficient is well-treated in Asmussen & Glynn [1, Chapter V]. The pedagogical point about shape matching versus correlation, while well-known to practitioners, does not appear to be highlighted in any of the standard textbook sources we consulted.

References

[1] S. Asmussen and P. W. Glynn, *Stochastic Simulation: Algorithms and Analysis*, Springer, 2007.

[2] P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, 2004.

[3] J. C. Hull, *Options, Futures, and Other Derivatives*, 10th edition, Pearson, 2018.